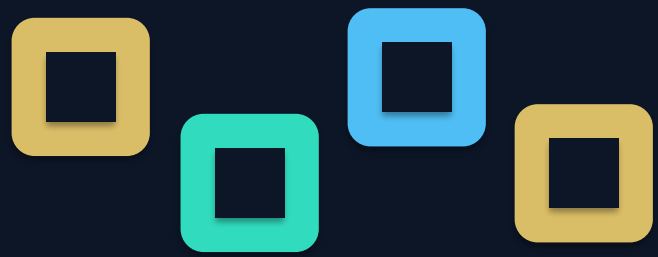
SDC HACKATHON 2026 | DOMAIN 1

TrendPilot AI

The Aiversecity Future-Ready Workforce Domain

Connect today’s skills to tomorrow’s career possibilities.



TEAM 404 BRAIN NOT FOUND

Team members: Teng Ray Tzong (Leader)

Loh Haw Cherng

A human-centred career exploration prototype

01 / PROBLEM

AI is changing work faster than people can adapt

The Domain 1 brief frames reskilling as a moving target—not a one-time career quiz.

56%

of Malaysian professionals use AI tools at work

26%

have received formal AI training

600k+

workers may need reskilling within 3–5 years

Source: Domain 1 brief, Background (p. 2); the brief attributes the first two figures to Hays Asia Salary Guide 2025 and the reskilling estimate to MOSTI.

Workers

Which skills should I learn next?

Employers

Which roles should we reskill for?

Graduates

Where do my current skills lead?



Start with people facing a career decision

The solution is designed to support exploration and reskilling—not make hiring decisions.

Working adults at risk of displacement

Need to identify transferable skills and practical next steps.

Fresh graduates & career switchers

Need to compare possible roles and understand skill gaps.

HR & learning-and-development teams

Need a clearer way to think about workforce reskilling priorities.

University & TVET career centres

Need an accessible starting point for career guidance.

Core user need: move from “What skills do I have?” to “What could I learn next—and why?”



From current skills to a future-ready pathway

This journey follows the workflow described in the Domain 1 brief (p. 3).

01

PROFILE

Enter skills and career interests

→

02

TREND SIGNALS

Review sourced market and AI signals

→

03

MATCH

Compare future roles and skill gaps

→

04

RESKILL

Follow an ordered learning pathway

→

05

SDG LINK

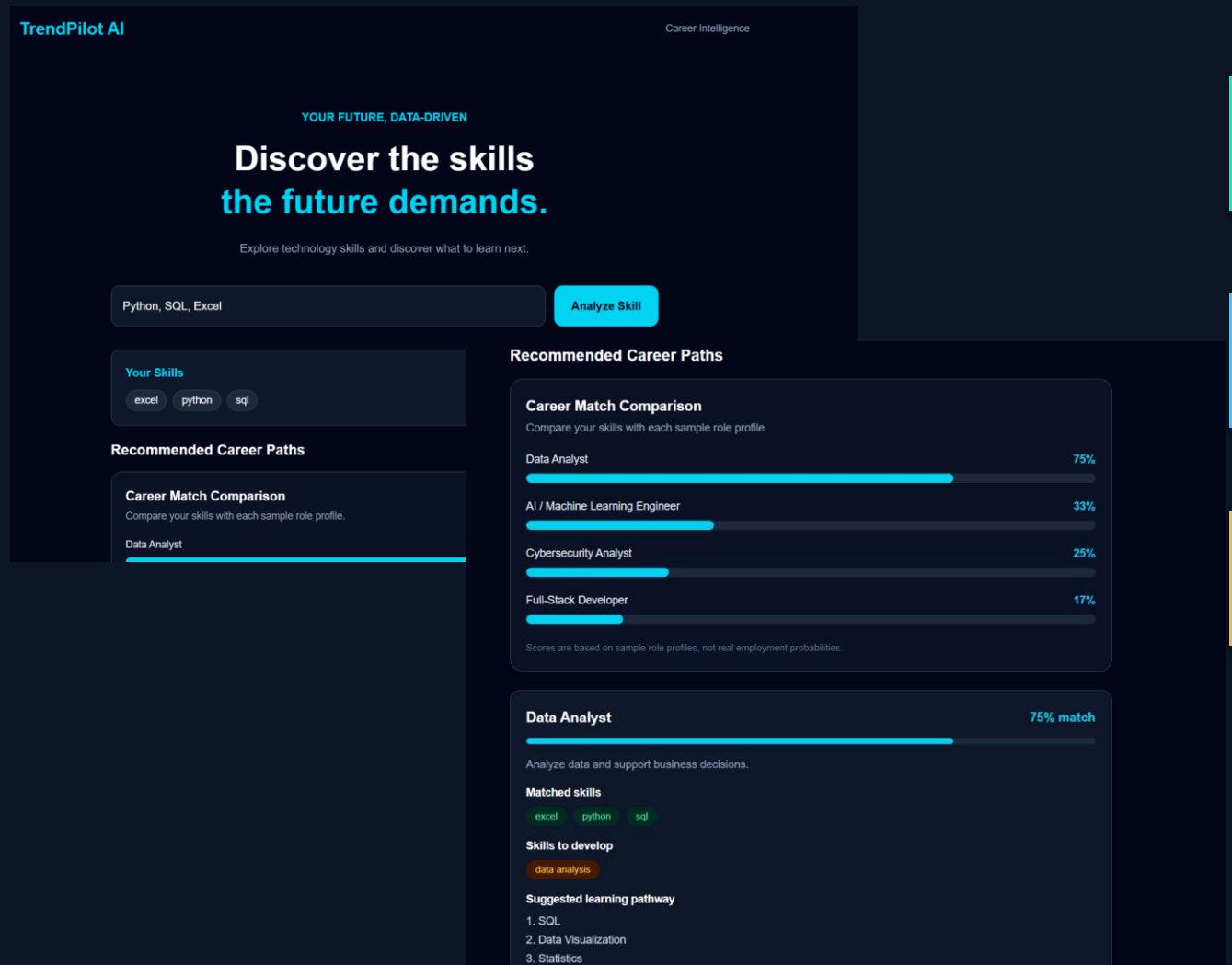
Understand the wider contribution

Design principle: every recommendation should explain its reasoning, show sources, and leave the final decision to the user.



A working starting point: skills in, career options out

The uploaded code includes a skill input, AI-generated role suggestions, match estimates, skill gaps, learning steps and SDG alignment.



Current features in code

Skill input • Groq-hosted LLM analysis • ranked role suggestions

Career guidance details

Matched skills • skills to develop • learning pathway • SDG alignment

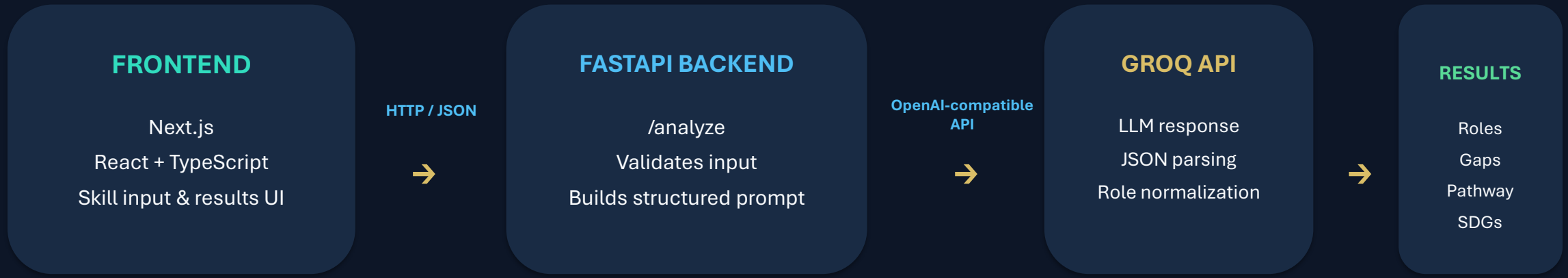
Employment snapshot

DOSM-labelled indicators with report date, source URL and limitations

Note: market figures are hard-coded snapshot values in the current code; they are not a live or auto-updating feed.

A simple web stack with an LLM-backed analysis endpoint

Architecture is based on the uploaded source code; keep the API key in a local environment file.



Market endpoint: /market-trends | Career endpoint: POST /analyze | Secret: GROQ_API_KEY (never commit)

How the prototype maps to the six required features

The Domain 1 brief lists six must-have features (pp. 4–5). Status below is based on the uploaded code, not a claim of full compliance.

1. Skills profile intake	Text-based skill input	IN CODE
2. AI-trend signal layer	DOSM-labelled static snapshot; needs an updatable feed	PARTIAL
3. Future-role matching	LLM-generated roles, match estimates and skill gaps	IN CODE
4. Personalised reskilling	Ordered learning steps are returned by the LLM	IN CODE
5. SDG mapping	Role-level SDG alignment is returned by the LLM	IN CODE
6. Handling uncertainty	Multiple scenarios are not implemented yet	NEXT

Priority gap for judging: make the trend-signal layer genuinely updateable and show at least two explicit future scenarios.



The hardest risk is trustworthy, updateable trend data

The brief explicitly says hard-coded lists alone do not satisfy the AI-trend signal requirement (p. 4).

Current baseline

DOSM-labelled employment indicators include a source URL, release date and limitation note—but are currently hard-coded.

Risk

National vacancy totals are not technology-specific signals and do not directly prove a particular skill is rising or declining.

Fallback plan

Add a dated public dataset or feed; store source + retrieval date; if refresh fails, clearly label the last successful snapshot.

TWO-SCENARIO DESIGN (PROPOSED)

MODERATE AI ADOPTION

Skills shift gradually; learning plan focuses on adjacent skills.

FAST AI DISRUPTION

Roles change faster; learning plan prioritizes adaptable skills.

Scenarios are proposed examples—not forecasts produced by the current app.

Career guidance should empower people—not decide for them

The brief highlights SDG 4 as the main SDG and SDGs 5, 8, 9 and 10 as relevant (p. 1).

SDG 4

Quality Education

Reskilling and lifelong learning

SDG 5

Gender Equality

Fair access to future skills

SDG 8

Decent Work

Productive employment

SDG 9

Innovation

Technology and infrastructure

SDG 10

Reduced Inequalities

More inclusive opportunity

Transparency

Show why a role was suggested, what is missing, and where external data came from.

Human control

Match scores are estimates—not employment probabilities or hiring decisions.

Demo data should be public, open or synthetic. Never upload real CVs or personal data.

09 / NEXT STEPS

A focused path from prototype to future-ready guidance

Build the missing evidence before adding more features.

NOW — Validate

- Test a few sample profiles
- Verify model output and error handling
- Show the live skill-to-role flow

NEXT — Strengthen signals

- Connect an updateable public source
- Display source date in the UI
- Compare moderate vs fast scenarios

THEN — Evaluate

- Check recommendation usefulness
- Audit source accuracy and SDG links
- Gather feedback from target users

TrendPilot AI

Connect skills. Explore futures. Keep people in control.

Sources & implementation notes

Keep this slide for Q&A; it does not need to be presented unless asked.

Domain requirements

SDC Hackathon 2026, Domain 1: The Aiversecity Future-Ready Workforce Domain — Problem Statement & Marking Rubric, pp. 1–7.

Employment indicator reference

Department of Statistics Malaysia (DOSM), Employment Statistics, First Quarter 2026. URL in the current backend code; figures are hard-coded snapshot values.

AI analysis implementation

The uploaded backend uses the OpenAI-compatible client configured for the Groq API and expects GROQ_API_KEY in backend/.env.

Disclosure

Role scores are model-generated estimates. Current code does not yet implement an updateable trend feed or explicit scenario comparison.

Before presenting: replace the illustrative UI with a real screenshot, verify all displayed outputs, and add the correct GitHub/demo links.